Social learning by type-token frequency - analysis

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Abstract

Abstract (to be written). Discuss quorum sensing [Sumpter and Pratt, 2009]

We excluded 0 participants because they failed the attention test.

We next analyze the SD and combine the results with the data frame for the main experiment.

1 Summary

1.1 Ratings

1.1.1 Prevalence

We observed a main effect of valency, suggesting that good actions were judged better than neutral actions, which were judged better than neutral actions. VERIFY THIS. GLMER: good > neutral = bad

We also observed a main effect of Action Status, suggesting that single occurence actions were judged as less prevalent than low frequency actions, which we judged as less prevalent than high type frequency actions.

The interaction was not significant.

1.1.2 Third party acceptability

We observed a main effect of valency, suggesting that good actions were judged better than neutral actions, which were judged better than neutral actions.

We also observed a main effect of Action Status, suggesting that single occurence actions were judged as worse than low frequency actions, which we judged as worse than high type frequency actions.

These factors did not interact.

1.1.3 First party acceptability

We observed a main effect of valency, suggesting that good actions were judged better than neutral actions, which were judged better than neutral actions. VERIFY THIS. GLMER: good > neutral = bad

We also observed a main effect of Action Status, suggesting that single occurence actions were judged as worse than low frequency actions, which we judged as worse than high type frequency actions.

These factors interacted. The effect of Action Status was only significant for neutral and good action, where the effect size for neutral actions was twice as high as for good actions.

1.2 Difference scores

1.2.1 Prevalence

The difference score comparing the high/low type frequency condition against the single occurrence condition yielded a significant main effect of Action Status, suggesting that the difference score was greater for the high type frequency condition than for the low type frequency condition. While the main effect of valency did not reach significance, the the interaction reached significance. The effect size associated with Action Status was much larger for neutral scenarios.

The difference score comparing the high and the low typ frequency condition revealed a significant main effect of Valency, suggesting that that the difference score did not differ between bad and neutral actions, but between neutral and good actions.

1.2.2 Third party acceptability

The difference score comparing the high/low type frequency condition against the single occurrence condition yielded a significant main effect of Action Status, suggesting that the difference score was greater for the high type frequency condition than for the low type frequency condition. Neither the main effect of valency nor the interaction reached significance.

Following up this interaction, we found that the main effect of Valency was only significant for high type frequency actions, but not for low frequency actions. Further, the effect size association with Action Status was much larger for neutral scenarios.

The difference score comparing the high and the low typ frequency condition revealed a marginal main effect of Valency, suggesting that that the difference score did not differ between bad and neutral actions, but differed marginally between neutral and good actions.

1.2.3 First party acceptability

The difference score comparing the high/low type frequency condition against the single occurrence condition yielded a significant main effect of Action Status, suggesting that the difference score was greater for the high type frequency condition than for the low type frequency condition. The main effect of valency reached significance, as did the interaction.

Following up this interaction, we found that the main effect of Valency was only significant for high type frequency actions, but not for low frequency actions. Further, the effect size association with Action Status was much larger for neutral scenarios, and reached significance only for these scenarios.

The difference score comparing the high and the low typ frequency condition revealed a marginal main effect of Valency, suggesting that that the difference score differed marginally between bad and neutral actions, but differed significantly between neutral and good actions.

2 Raw ratings

2.1 Ratings (analyzed using ANOVAs)

Table 1: Descriptive for the untransformed data. The p value reflects a Wilcoxon test against the scale midpoint of 2.5.

	N	M	SE	Cohen's d	p
bad - prevalence					
highTypeFreq	17	3.53	0.104	8.45	0.000
lowTypeFreq	17	2.75	0.164	4.18	0.181
single	17	2.09	0.115	4.53	0.005
neutral - prevalence					
highTypeFreq	17	3.51	0.117	7.48	0.000
lowTypeFreq	17	2.50	0.091	6.88	0.875
single	17	1.96	0.114	4.29	0.001
good - prevalence					
highTypeFreq	18	3.75	0.071	12.91	0.000
lowTypeFreq	18	3.29	0.134	5.97	0.000
single	18	2.70	0.149	4.39	0.303
bad - third.party.acceptability					
highTypeFreq	17	2.71	0.146	4.63	0.192
lowTypeFreq	17	1.89	0.121	3.88	0.001
single	17	1.66	0.115	3.59	0.000
neutral - third.party.acceptability					
highTypeFreq	17	3.46	0.095	9.14	0.000
lowTypeFreq	17	2.72	0.098	6.92	0.038
single	17	2.32	0.119	4.89	0.097
good - third.party.acceptability					
highTypeFreq	18	3.80	0.060	15.42	0.000
lowTypeFreq	18	3.45	0.117	7.16	0.000
single	18	3.00	0.155	4.70	0.007
bad - first.party.acceptablity					
highTypeFreq	17	1.61	0.094	4.30	0.000
lowTypeFreq	17	1.57	0.099	3.98	0.000
single	17	1.54	0.097	4.00	0.000
neutral - first.party.acceptablity					
highTypeFreq	17	2.81	0.106	6.64	0.004
lowTypeFreq	17	2.59	0.105	6.19	0.705
single	17	2.47	0.124	4.96	0.549
good - first.party.acceptablity					
highTypeFreq	18	3.69	0.070	12.86	0.000
lowTypeFreq	18	3.68	0.075	11.85	0.000
single	18	3.63	0.083	10.57	0.000

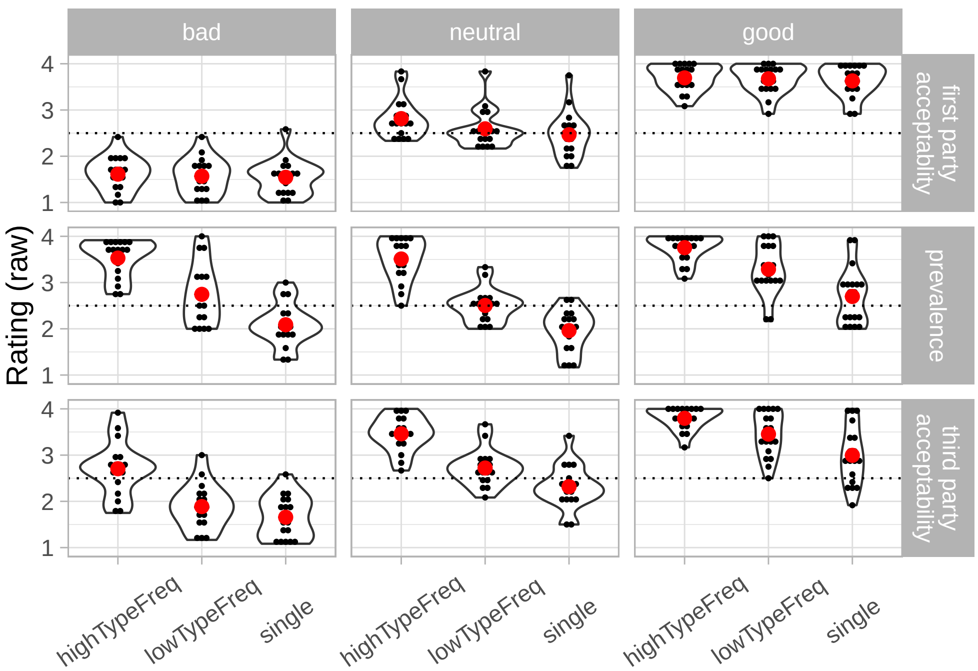

Figure 1: Raw ratings

Table 2: SW acceptability for participants themselves

myValence	myQueriedActionStatus	W	p.value	p<=.05
bad	highTypeFreq	0.815	0.003	**
neutral	highTypeFreq	0.904	0.079	.
good	highTypeFreq	0.825	0.003	**
bad	lowTypeFreq	0.910	0.101	
neutral	lowTypeFreq	0.915	0.123	
good	lowTypeFreq	0.906	0.074	.
bad	single	0.954	0.519	
neutral	single	0.938	0.294	
good	single	0.883	0.029	*

2.1.1 Prevalence

As shown in Table 2, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 3, we observed the expected main effect of Valency on first-party acceptability. Further, we observer a main effect of Action Status. The interaction between Action Status and Valency did not reach significance.

As shown in Table 3, separate ANOVAs for each valency revealed a strong main effect of Action Status for all valencies.

Finally, Table 3 shows separate two way ANOVAs leaving out one of the Action Statuses. In all ANOVAs, we observed the expected main effect of Valency. We also observed a main effect of Action Status for all comparisons, suggesting that both an increased token frequency and an increased type frequency affect first party acceptability.

Finally, the interaction between Action Status and Valency failed to reach significance .

Table 3: ANOVA for prevalence judgments and unprocessed ratings. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

Effect	df1	df2	F	p	η_p^2
Overall					
myValence	2	49	18.200	0.000	0.426
myQueriedActionStatus	2	98	107.000	0.000	0.670
myQueriedActionStatus:myValence	4	98	1.920	0.114	0.024
Follow-up of valency: bad					
myQueriedActionStatus	2	32	31.800	0.000	0.665
Follow-up of valency: neutral					
myQueriedActionStatus	2	32	47.300	0.000	0.747
Follow-up of valency: good					
myQueriedActionStatus	2	34	31.200	0.000	0.647
Follow-up of Action Status: lowTypeFreq vs. highTypeFreq					
myValence	2	49	11.600	0.000	0.321
myQueriedActionStatus	1	49	62.700	0.000	0.535
myQueriedActionStatus:myValence	2	49	2.770	0.073	0.047
Follow-up of Action Status: single vs. highTypeFreq					
myValence	2	49	16.500	0.000	0.402
myQueriedActionStatus	1	49	170.000	0.000	0.761
myQueriedActionStatus:myValence	2	49	2.150	0.128	0.019
Follow-up of Action Status: single vs. lowTypeFreq					
myValence	2	49	13.600	0.000	0.357
myQueriedActionStatus	1	49	61.300	0.000	0.554
myQueriedActionStatus:myValence	2	49	0.182	0.834	0.003

Table 4: SW acceptability for participants themselves

myValence	myQueriedActionStatus	W	p.value	p<=.05
bad	highTypeFreq	0.958	0.598	
neutral	highTypeFreq	0.949	0.443	
good	highTypeFreq	0.823	0.003	**
bad	lowTypeFreq	0.967	0.762	
neutral	lowTypeFreq	0.938	0.294	
good	lowTypeFreq	0.913	0.099	.
bad	single	0.923	0.169	
neutral	single	0.949	0.439	
good	single	0.949	0.403	

2.1.2 Third-party acceptability

As shown in Table 4, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 5, we observed the expected main effect of Valency on third-party acceptability. Further, we observed a main effect of Action Status. The interaction between Action Status and Valency was not significant.

As shown in Table 5, separate ANOVAs for each valency revealed a strong main effect of Action Status for all scenarios, though the effect size for neutral actions was somewhat stronger.

Finally, Table 5 shows separate two way ANOVAs leaving out one of the Action Statuses. In all ANOVAs, we observed the expected main effect of Valency. We also observed a main effect of Action Status for all comparisons, suggesting that both an increased token frequency and an increased type frequency affect first party acceptability.

Finally, the interactions between Action Status and Valency failed to reach significance.

Table 5: ANOVA for third-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

Effect	df1	df2	F	p	η_p^2
Overall					
myValence	2	49	72.00	0.000	0.746
myQueriedActionStatus	2	98	75.50	0.000	0.589
myQueriedActionStatus:myValence	4	98	1.86	0.124	0.029
Follow-up of valency: bad					
myQueriedActionStatus	2	32	20.70	0.000	0.565
Follow-up of valency: neutral					
myQueriedActionStatus	2	32	45.50	0.000	0.740
Follow-up of valency: good					
myQueriedActionStatus	2	34	19.50	0.000	0.535
Follow-up of Action Status: lowTypeFreq vs. highTypeFreq					
myValence	2	49	75.90	0.000	0.756
myQueriedActionStatus	1	49	56.30	0.000	0.504
myQueriedActionStatus:myValence	2	49	3.15	0.052	0.056
Follow-up of Action Status: single vs. highTypeFreq					
myValence	2	49	62.40	0.000	0.718
myQueriedActionStatus	1	49	100.00	0.000	0.663
myQueriedActionStatus:myValence	2	49	1.07	0.350	0.014
Follow-up of Action Status: single vs. lowTypeFreq					
myValence	2	49	45.10	0.000	0.648
myQueriedActionStatus	1	49	41.40	0.000	0.444
myQueriedActionStatus:myValence	2	49	1.46	0.243	0.031

Table 6: SW acceptability for participants themselves

myValence	myQueriedActionStatus	W	p.value	p<=.05
bad	highTypeFreq	0.968	0.776	
neutral	highTypeFreq	0.864	0.018	*
good	highTypeFreq	0.901	0.061	.
bad	lowTypeFreq	0.967	0.758	
neutral	lowTypeFreq	0.824	0.004	**
good	lowTypeFreq	0.888	0.035	*
bad	single	0.911	0.103	
neutral	single	0.935	0.267	
good	single	0.889	0.037	*

2.1.3 First-party acceptability

As shown in Table 6, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 7, we observed the expected main effect of Valency on first-party acceptability. Further, we observerd a main effect of Action Status as well as an interaction between Action Status and Valency.

As shown in Table 7, separate ANOVAs for each valency revealed a strong main effect of Action Status for neutral scenarios, a weaker one for good scenarios, and none for bad scenarios. This is also expected, as the moral acceptability of neutral scenarios should be more malleable.

Finally, Table 7 shows separate two way ANOVAs leaving out one of the Action Statuses. In all ANOVAs, we observed the expected main effect of Valency. We also observed a main effect of Action Status for all comparisons, suggesting that both an increased token frequency and an increased type frequency affect first party acceptability.

Finally, the interaction between Action Status and Valency failed to reach significance only when single occurrence actions were compare to low type frequency actions, but was significant when single occurrence actions were compared to high Type Frequency actions and when the other two frequency conditions were compared. We further investigate this by simultaneous restricting the Action Status and the Valency.

Table 7: ANOVA for first-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

Effect	df1	df2	F	p	η_p^2
Overall					
myValence	2	49	146.00	0.000	0.856
myQueriedActionStatus	2	98	12.70	0.000	0.179
myQueriedActionStatus:myValence	4	98	4.55	0.002	0.129
Follow-up of valency: bad					
myQueriedActionStatus	2	32	2.12	0.137	0.117
Follow-up of valency: neutral					
myQueriedActionStatus	2	32	8.22	0.001	0.339
Follow-up of valency: good					
myQueriedActionStatus	2	34	3.56	0.039	0.173
Follow-up of Action Status: lowTypeFreq vs. highTypeFreq					
myValence	2	49	152.00	0.000	0.861
myQueriedActionStatus	1	49	10.30	0.002	0.149
myQueriedActionStatus:myValence	2	49	4.76	0.013	0.138
Follow-up of Action Status: single vs. highTypeFreq					
myValence	2	49	144.00	0.000	0.855
myQueriedActionStatus	1	49	16.30	0.000	0.212
myQueriedActionStatus:myValence	2	49	5.63	0.006	0.147
Follow-up of Action Status: single vs. lowTypeFreq					
myValence	2	49	130.00	0.000	0.842
myQueriedActionStatus	1	49	7.00	0.011	0.118
myQueriedActionStatus:myValence	2	49	1.57	0.219	0.053

Table 8: Differences for each frequency conditions separately for first person acceptability for all participants.

effect	df1	df2	F.value	p.value	partial.eta.squared
bad - lowTypeFreq vs. highTypeFreq					
myQueriedActionStatus	1	16	1.580	0.227	0.090
bad - single vs. highTypeFreq					
myQueriedActionStatus	1	16	5.060	0.039	0.240
bad - single vs. lowTypeFreq					
myQueriedActionStatus	1	16	0.475	0.501	0.029
neutral - lowTypeFreq vs. highTypeFreq					
myQueriedActionStatus	1	16	7.810	0.013	0.328
neutral - single vs. highTypeFreq					
myQueriedActionStatus	1	16	9.290	0.008	0.367
neutral - single vs. lowTypeFreq					
myQueriedActionStatus	1	16	4.810	0.043	0.231
good - lowTypeFreq vs. highTypeFreq					
myQueriedActionStatus	1	17	0.883	0.361	0.049
good - single vs. highTypeFreq					
myQueriedActionStatus	1	17	8.770	0.009	0.340
good - single vs. lowTypeFreq					
myQueriedActionStatus	1	17	2.130	0.163	0.111

2.2 Difference scores

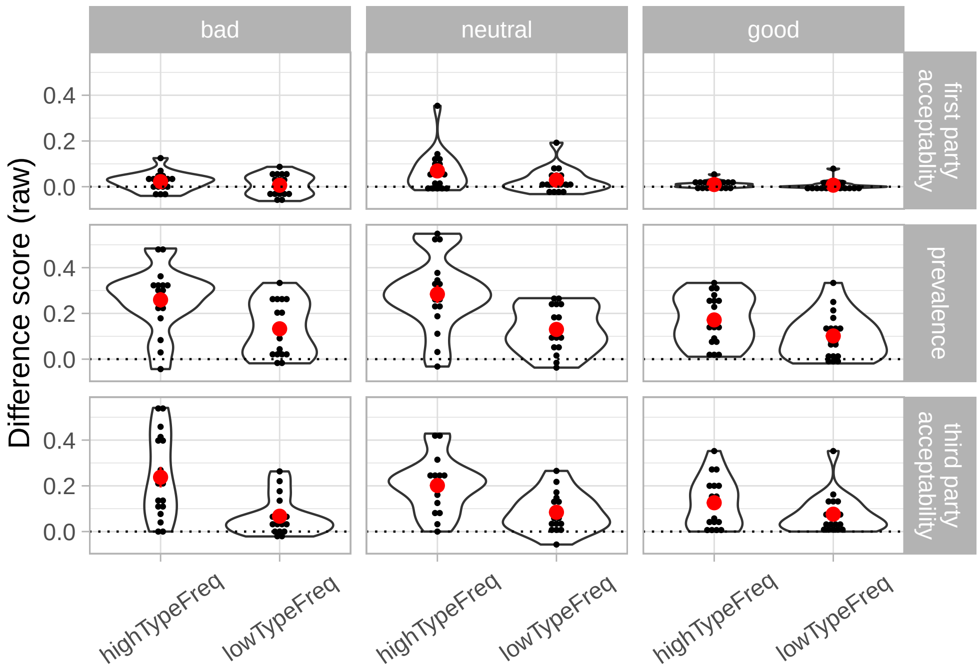

Figure 2: Raw difference scores vs. single occurrence actions

Table 9: Descriptive for the difference scores against the single occurrence condition using the untransformed data. The p value reflects a Wilcoxon test against zero.

Valency	Action Status	N	M	SE	Cohen's d	p
first party acceptablity						
bad	highTypeFreq	17	0.022	0.010	0.551	0.054
bad	lowTypeFreq	17	0.007	0.011	0.147	0.551
neutral	highTypeFreq	17	0.069	0.022	0.776	0.003
neutral	lowTypeFreq	17	0.029	0.014	0.530	0.045
good	highTypeFreq	18	0.010	0.004	0.659	0.014
good	lowTypeFreq	18	0.007	0.005	0.330	0.213
prevalence						
bad	highTypeFreq	17	0.259	0.035	1.856	0.000
bad	lowTypeFreq	17	0.133	0.030	1.120	0.001
neutral	highTypeFreq	17	0.284	0.040	1.773	0.000
neutral	lowTypeFreq	17	0.130	0.025	1.299	0.001
good	highTypeFreq	18	0.172	0.026	1.580	0.000
good	lowTypeFreq	18	0.102	0.024	1.031	0.001
third party acceptability						
bad	highTypeFreq	17	0.238	0.046	1.286	0.001
bad	lowTypeFreq	17	0.067	0.021	0.802	0.002
neutral	highTypeFreq	17	0.201	0.029	1.713	0.000
neutral	lowTypeFreq	17	0.084	0.021	0.997	0.002
good	highTypeFreq	18	0.126	0.026	1.171	0.000
good	lowTypeFreq	18	0.076	0.021	0.889	0.000

Table 11: SW acceptability for participants themselves

myValence	myQueriedActionStatus	W	p.value	p<=.05
bad	highTypeFreq	0.939	0.307	
neutral	highTypeFreq	0.955	0.538	
good	highTypeFreq	0.929	0.190	
bad	lowTypeFreq	0.900	0.067	.
neutral	lowTypeFreq	0.936	0.269	
good	lowTypeFreq	0.929	0.187	

Table 9: Descriptive for the difference scores against the single occurrence condition using the untransformed data. The p value reflects a Wilcoxon test against zero. *(continued)*

Valency	Action Status	N	M	SE	Cohen's d	p

Table 10: Descriptive for the difference scores against the single occurrence condition using the untransformed data. The p value reflects a Wilcoxon test against zero.

Valency	N	M	SE	Cohen's d	p
first party acceptablity					
bad	17	0.022	0.010	0.551	0.054
neutral	17	0.069	0.022	0.776	0.003
good	18	0.010	0.004	0.659	0.014
prevalence					
bad	17	0.259	0.035	1.856	0.000
neutral	17	0.284	0.040	1.773	0.000
good	18	0.172	0.026	1.580	0.000
third party acceptability					
bad	17	0.238	0.046	1.286	0.001
neutral	17	0.201	0.029	1.713	0.000
good	18	0.126	0.026	1.171	0.000

2.2.1 Prevalence

As shown in Table 11, no of cells deviated from normality.

As shown in Table 12, we observed no main effect of Valency on the difference score against single occurrence actions for prevalence. We observed a main effect of Action Status as well as a marginal interaction between Action Status and Valency.

As shown in Table 12, separate ANOVAs for each valency revealed a strong main effect of Action Status only for all three valencies.

Finally, Table 12 shows separate ANOVAs for each Action Status. The main effect of valency was significant only for high type frequency scenarios.

2.2.1.1 Difference scores comparing high type frequency against low type frequency As shown in Table 13, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

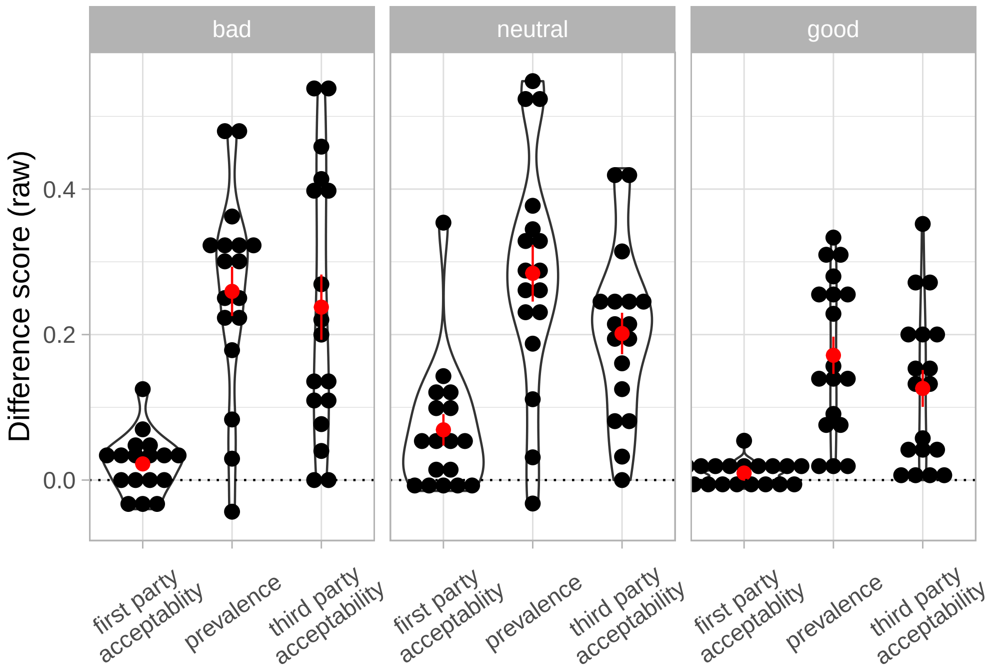

Figure 3: Raw difference scores high vs. low type frequency actions

Table 12: ANOVA for difference scores comparing each frequency condition to the single occurrence condition. first-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

Effect	df1	df2	F	p	η_p^2
Overall					
myValence	2	49	2.090	0.134	0.079
myQueriedActionStatus	1	49	60.800	0.000	0.527
myQueriedActionStatus:myValence	2	49	2.810	0.070	0.049
Follow-up of valency: bad					
myQueriedActionStatus	1	16	15.100	0.001	0.486
Follow-up of valency: neutral					
myQueriedActionStatus	1	16	40.000	0.000	0.714
Follow-up of valency: good					
myQueriedActionStatus	1	17	13.000	0.002	0.434
Follow-up of Action Status: highTypeFreq					
myValence	2	49	3.280	0.046	0.118
Follow-up of Action Status: lowTypeFreq					
myValence	2	49	0.469	0.628	0.019

As shown in Table 14, we observed a marginal main effect of Valency on difference scores for first-party acceptability.

As shown in Table 14, separate ANOVAs for each valency pair revealed a strong main effect of Action Status

Table 13: SW acceptability for participants themselves for high vs. low difference scores

myValence	W	p.value	p<=.05
bad	0.909861807460269	0.000912857569696844	***
neutral	0.944343656501613	0.018409467014131	*
good	0.842486132732877	4.92791902746997e-06	

only for the bad vs. good comparison the the neutral vs. good comparison.

Table 14: ANOVA for difference scores comparing the high vs low type frequency conditions for prevalence. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

myValence	Effect	df1	df2	F	p	η_p^2
Overall						
NA	myValence	2	49	3.280	0.046	0.118
By valency						
neutral vs. good	myValence	1	33	5.990	0.020	0.154
bad vs. good	myValence	1	33	4.340	0.045	0.116
bad vs. neutral	myValence	1	32	0.232	0.633	0.007

Table 15: SW acceptability for participants themselves

myValence	myQueriedActionStatus	W	p.value	p<=.05
bad	highTypeFreq	0.914	0.116	
neutral	highTypeFreq	0.961	0.659	
good	highTypeFreq	0.921	0.136	
bad	lowTypeFreq	0.858	0.014	*
neutral	lowTypeFreq	0.969	0.801	
good	lowTypeFreq	0.780	0.001	***

2.2.2 Third-party acceptability

2.2.2.1 Difference scores against single actions As shown in Table 15, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 16, we observed no main effect of Valency on the difference score against single occurrence actions for first-party acceptability. , we observed a main effect of Action Status as well as an interaction between Action Status and Valency.

As shown in Table 16, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Finally, Table 16 shows separate ANOVAs for each Action Status. The main effect of valency was marginal;ly significant only for high type frequency scenarios.

Table 16: ANOVA for difference scores comparing each frequency condition to the single occurrence condition. third-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

Effect	df1	df2	F	p	η_p^2
Overall					
myValence	2	49	1.280	0.288	0.050
myQueriedActionStatus	1	49	52.100	0.000	0.467
myQueriedActionStatus:myValence	2	49	5.170	0.009	0.093
Follow-up of valency: bad					
myQueriedActionStatus	1	16	18.500	0.001	0.537
Follow-up of valency: neutral					
myQueriedActionStatus	1	16	37.500	0.000	0.701
Follow-up of valency: good					
myQueriedActionStatus	1	17	9.520	0.007	0.359
Follow-up of Action Status: highTypeFreq					
myValence	2	49	2.900	0.064	0.106
Follow-up of Action Status: lowTypeFreq					
myValence	2	49	0.174	0.841	0.007

2.2.2.2 Difference scores comparing high type frequency against low type frequency As shown in Table 17, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

Table 17: SW acceptability for participants themselves for high vs. low difference scores

myValence	W	p.value	p<=.05
bad	0.909861807460269	0.000912857569696844	***
neutral	0.944343656501613	0.018409467014131	*
good	0.842486132732877	4.92791902746997e-06	

As shown in Table 18, we observed a marginal main effect of Valency on difference scores for first-party acceptability.

As shown in Table 18, separate ANOVAs for each valency pair revealed a strong main effect of Action Status only for the bad vs. good comparison, and a marginal one for the neutral vs. good comparison.

Table 18: ANOVA for difference scores comparing the high vs low type frequency conditions for first-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

myValence	Effect	df1	df2	F	p	η_p^2
Overall						
NA	myValence	2	49	2.900	0.064	0.106
By valency						
neutral vs. good	myValence	1	33	3.930	0.056	0.106
bad vs. good	myValence	1	33	4.850	0.035	0.128
bad vs. neutral	myValence	1	32	0.465	0.500	0.014

Table 19: SW acceptability for participants themselves

myValence	myQueriedActionStatus	W	p.value	p<=.05
bad	highTypeFreq	0.941	0.335	
neutral	highTypeFreq	0.776	0.001	***
good	highTypeFreq	0.821	0.003	**
bad	lowTypeFreq	0.946	0.391	
neutral	lowTypeFreq	0.831	0.006	**
good	lowTypeFreq	0.679	0.000	***

2.2.3 First-party acceptability

2.2.3.1 Difference scores against single actions As shown in Table 19, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

As shown in Table 20, we observed the expected main effect of Valency on first-party acceptability. Further, we observed a main effect of Action Status as well as an interaction between Action Status and Valency. (Not that including the factor Action Status is redundant with the direct difference score below.)

As shown in Table 20, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Finally, Table 20 shows separate ANOVAs for each Action Status. The main effect of valency was significant only for high type frequency scenarios.

Table 20: ANOVA for difference scores comparing each frequency condition to the single occurrence condition. first-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

Effect	df1	df2	F	p	η_p^2
Overall					
myValence	2	49	4.100	0.023	0.143
myQueriedActionStatus	1	49	10.400	0.002	0.158
myQueriedActionStatus:myValence	2	49	3.430	0.040	0.103
Follow-up of valency: bad					
myQueriedActionStatus	1	16	1.610	0.223	0.091
Follow-up of valency: neutral					
myQueriedActionStatus	1	16	9.450	0.007	0.371
Follow-up of valency: good					
myQueriedActionStatus	1	17	0.937	0.347	0.052
Follow-up of Action Status: highTypeFreq					
myValence	2	49	5.310	0.008	0.178
Follow-up of Action Status: lowTypeFreq					
myValence	2	49	1.560	0.220	0.060

2.2.3.2 Difference scores comparing high type frequency against low type frequency As shown in Table 21, a number of cells deviated from normality. TODO: TRY TO HEAL IT BY TRANSFORMING

Table 21: SW acceptability for participants themselves for high vs. low difference scores

myValence	W	p.value	p<=.05
bad	0.909861807460269	0.000912857569696844	***
neutral	0.944343656501613	0.018409467014131	*
good	0.842486132732877	4.92791902746997e-06	

As shown in Table 22, we observed the expected main effect of Valency on difference scores for first-party acceptability.

As shown in Table 22, separate ANOVAs for each valency revealed a strong main effect of Action Status only for neutral scenarios. The main effect was not significant for good or bad scenarios. Again, this is redundant with the direct difference scores.

Table 22: ANOVA for difference scores comparing the high vs low type frequency conditions for first-person acceptability. The data combines male and female participants. (Top) Overall ANOVA. (Middle) Differences between frequency conditions separately for each valency for first person acceptability for all participants. (Bottom). Differences for each frequency conditions separately for first person acceptability for all participants.

myValence	Effect	df1	df2	F	p	η_p^2
Overall						
NA	myValence	2	49	5.31	0.008	0.178
By valency						
neutral vs. good	myValence	1	33	7.81	0.009	0.191
bad vs. good	myValence	1	33	1.54	0.223	0.045
bad vs. neutral	myValence	1	32	3.90	0.057	0.109

Table 23: P values of significant correlations between dark triad predictors and difference scores comparing ratings to the equivalent ratings in the single occurrence condition,

var	Narcissism	Psychopathy
bad		
highTypeFreq - prevalence	0.001	0.012
highTypeFreq - third party acceptability	0.041	0.128

Table 24: P values of significant correlations between dark triad predictors and difference scores comparing comparing the high and the low type frequency conditions.

var	Narcissism	Psychopathy
bad		
prevalence	0.001	0.012
third party acceptability	0.041	0.128

3 Correlations

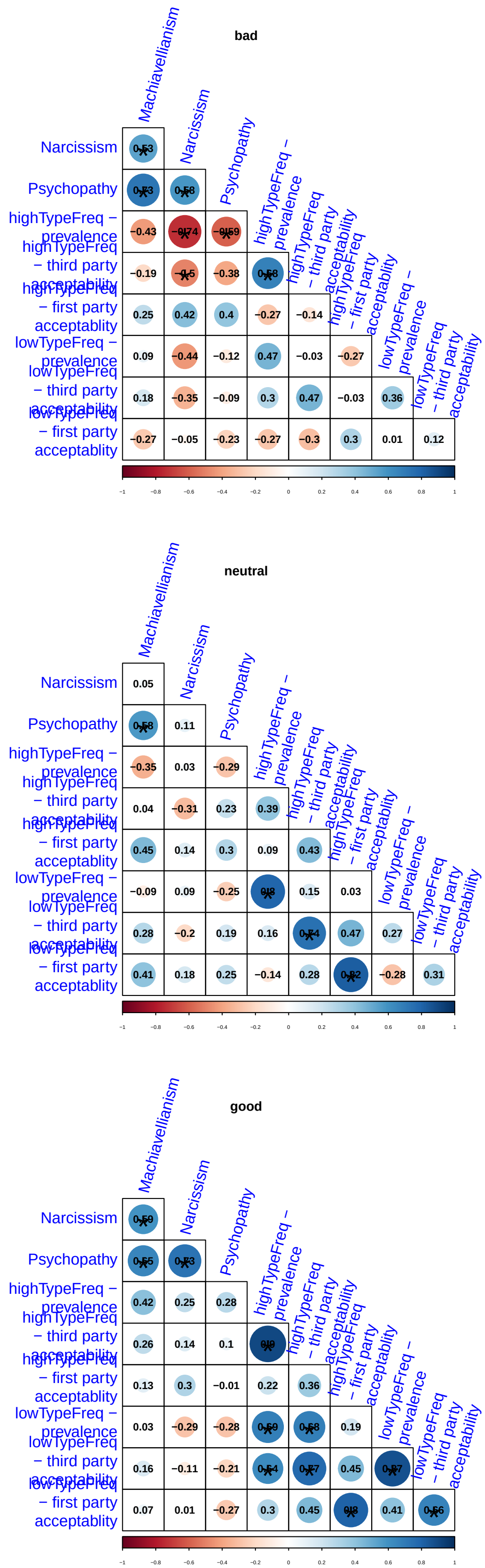

Figure 4: Correlation for difference scores comparing actions against single occurrence actions.

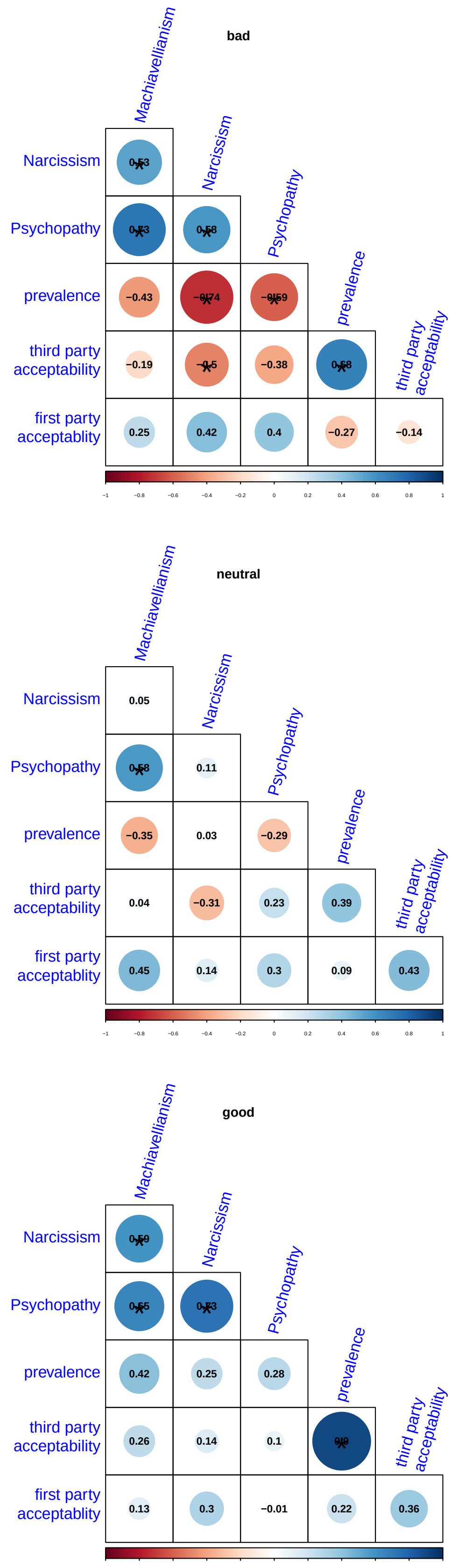

Figure 5: Correlation for difference scores comparing high vs. low type frequency actions.

4 Binary ratings (analyzed using GLMER)

4.1 Ratings

Table 25: Descriptives for the data into binary ratings.

	N	M	SE
bad - prevalence			
highTypeFreq	17	0.936	0.025
lowTypeFreq	17	0.534	0.104
single	17	0.235	0.073
neutral - prevalence			
highTypeFreq	17	0.882	0.040
lowTypeFreq	17	0.431	0.063
single	17	0.206	0.045
good - prevalence			
highTypeFreq	18	0.977	0.012
lowTypeFreq	18	0.819	0.066
single	18	0.546	0.086
bad - third.party.acceptability			
highTypeFreq	17	0.637	0.077
lowTypeFreq	17	0.211	0.061
single	17	0.103	0.039
neutral - third.party.acceptability			
highTypeFreq	17	0.882	0.037
lowTypeFreq	17	0.569	0.054
single	17	0.382	0.055
good - third.party.acceptability			
highTypeFreq	18	0.977	0.012
lowTypeFreq	18	0.894	0.041
single	18	0.676	0.078
bad - first.party.acceptablity			
highTypeFreq	17	0.123	0.030
lowTypeFreq	17	0.098	0.028
single	17	0.088	0.036
neutral - first.party.acceptablity			
highTypeFreq	17	0.603	0.050
lowTypeFreq	17	0.539	0.047
single	17	0.500	0.050
good - first.party.acceptablity			
highTypeFreq	18	0.940	0.022
lowTypeFreq	18	0.944	0.023
single	18	0.931	0.024

4.1.1 Prevalence

4.1.2 Third-party acceptability

4.1.3 First-party acceptability

4.2 Difference scores (GLMER or ANOVA?)

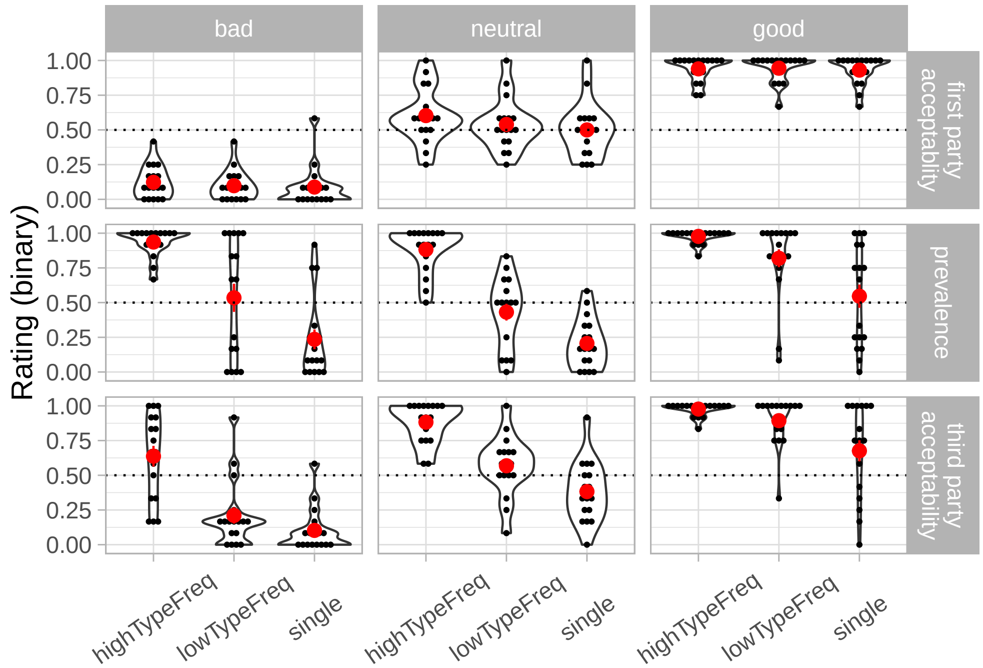

Figure 6: Bin ratings

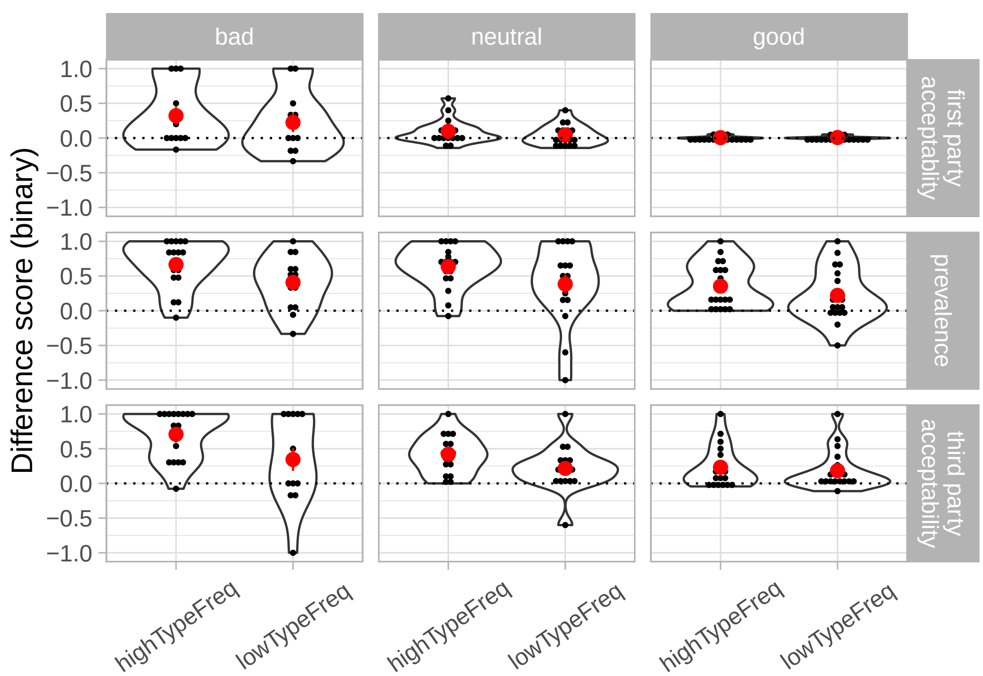

Figure 7: Binary difference scores vs. single occurrence actions

Table 26: GLMER results, using the full models.

term	Log-odds Estimate	SE	CI	Odd ratios Estimate	SE	CI	t	p
Prevalence								
(Intercept)	-1.522	0.376	[-2.26, -0.785]	0.218	0.082	[0.104, 0.456]	-4.046	0.000
myQueriedActionStatuslowTypeFreq	1.203	0.238	[0.737, 1.67]	3.329	0.791	[2.09, 5.3]	5.062	0.000
myQueriedActionStatushighTypeFreq	3.766	0.306	[3.17, 4.36]	43.212	13.203	[23.7, 78.6]	12.326	0.000
myValencebad	-0.122	0.539	[-1.18, 0.934]	0.885	0.477	[0.308, 2.54]	-0.227	0.821
myValencegood	1.864	0.524	[0.838, 2.89]	6.450	3.378	[2.31, 18]	3.560	0.000
myQueriedActionStatuslowTypeFreq:myValencebad	0.643	0.359	[-0.0602, 1.35]	1.902	0.682	[0.942, 3.84]	1.792	0.073
myQueriedActionStatushighTypeFreq:myValencebad	1.335	0.513	[0.33, 2.34]	3.800	1.949	[1.39, 10.4]	2.603	0.009
myQueriedActionStatuslowTypeFreq:myValencegood	0.632	0.360	[-0.0732, 1.34]	1.882	0.677	[0.929, 3.81]	1.757	0.079
myQueriedActionStatushighTypeFreq:myValencegood	0.625	0.598	[-0.547, 1.8]	1.869	1.118	[0.579, 6.04]	1.045	0.296
Third party acceptability								
(Intercept)	-0.540	0.337	[-1.2, 0.122]	0.583	0.197	[0.301, 1.13]	-1.599	0.110
myQueriedActionStatuslowTypeFreq	0.906	0.222	[0.47, 1.34]	2.473	0.549	[1.6, 3.82]	4.076	0.000
myQueriedActionStatushighTypeFreq	2.876	0.286	[2.32, 3.44]	17.743	5.076	[10.1, 31.1]	10.052	0.000
myValencebad	-2.046	0.517	[-3.06, -1.03]	0.129	0.067	[0.047, 0.356]	-3.959	0.000
myValencegood	1.583	0.481	[0.64, 2.53]	4.872	2.344	[1.9, 12.5]	3.290	0.001
myQueriedActionStatuslowTypeFreq:myValencebad	0.061	0.378	[-0.679, 0.801]	1.063	0.401	[0.507, 2.23]	0.161	0.872
myQueriedActionStatushighTypeFreq:myValencebad	0.409	0.419	[-0.412, 1.23]	1.506	0.631	[0.662, 3.42]	0.977	0.329
myQueriedActionStatuslowTypeFreq:myValencegood	0.831	0.368	[0.11, 1.55]	2.297	0.846	[1.12, 4.73]	2.258	0.024
myQueriedActionStatushighTypeFreq:myValencegood	0.610	0.568	[-0.502, 1.72]	1.841	1.045	[0.605, 5.6]	1.076	0.282
First party acceptability								
(Intercept)	0.151	0.721	[-1.26, 1.56]	1.163	0.838	[0.283, 4.78]	0.209	0.834
myQueriedActionStatuslowTypeFreq	0.302	0.275	[-0.238, 0.842]	1.353	0.373	[0.788, 2.32]	1.097	0.273
myQueriedActionStatushighTypeFreq	0.793	0.279	[0.246, 1.34]	2.211	0.617	[1.28, 3.82]	2.843	0.004
myValencebad	-4.087	1.082	[-6.21, -1.97]	0.017	0.018	[0.00201, 0.14]	-3.778	0.000
myValencegood	4.853	1.141	[2.62, 7.09]	128.135	146.241	[13.7, 1200]	4.252	0.000
myQueriedActionStatuslowTypeFreq:myValencebad	-0.119	0.510	[-1.12, 0.88]	0.888	0.453	[0.327, 2.41]	-0.233	0.816
myQueriedActionStatushighTypeFreq:myValencebad	-0.207	0.499	[-1.18, 0.771]	0.813	0.406	[0.306, 2.16]	-0.414	0.679
myQueriedActionStatuslowTypeFreq:myValencegood	0.052	0.560	[-1.05, 1.15]	1.053	0.590	[0.351, 3.16]	0.093	0.926
myQueriedActionStatushighTypeFreq:myValencegood	-0.564	0.555	[-1.65, 0.523]	0.569	0.316	[0.192, 1.69]	-1.017	0.309

4.2.1 Prevalence

4.2.2 Third-party acceptability

4.2.3 First-party acceptability

References

David J. T. Sumpter and Stephen C. Pratt. Quorum responses and consensus decision making. *Philosophical transactions of the Royal Society of London. Series B, Biological sciences*, 364:743–753, March 2009. ISSN 1471-2970. doi: 10.1098/rstb.2008.0204.

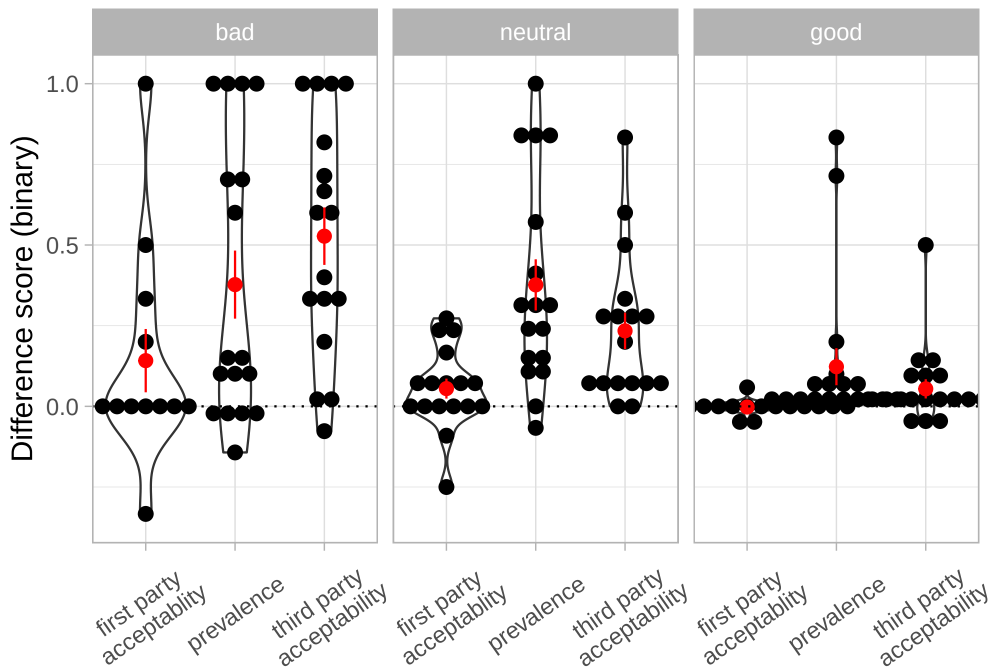

Figure 8: Binary difference scores high vs. low type frequency actions